Lunar Greenhouse

Campaign 1 · Case 02 · Shackleton Crater, Lunar South Pole

SOLAR AGRICULTURAL AGENCY
Teacher Guide

ONE-PERIOD PURPOSE

Students act as Process Modelers: reconstruct the tomato reproductive sequence, locate the first failed step, reject distractors, write CER, and design a verifiable support.

Distribute

- Student or Accessible mission packet
- Game-capable device; headphones optional
- Fallback evidence digest if needed

Correct diagnosis

Missing effective flower agitation leaves pollen on anthers; stigma contact and later fruit-set events do not occur.

Collect

Completed mission packet. Never use game score, speed, rank, rapport, or bonus exploration as academic evidence.

Likely sticking point

“Self-pollinating” does not mean no pollen movement. Require the sequence and undisturbed-pollen evidence, not merely “no bees.”

Time	Action
0–5	Open game and packet.
5–10	Launch flower-without-fruit phenomenon; **2 · Initial thinking**.
10–28	One formal clue from Crew, Sensors, Plants, Logs.
28–38	**3 · Model the pollination sequence** and **4 · Identify the failed step**.
38–48	**5 · Test the competing explanations**, **6 · Diagnose and reject an alternative**, and **7 · Claim-Evidence-Reasoning**.
48–55	**8 · Design a reliable pollination support**.
55–60	**9 · Exit ticket** independently.

Fallback: Give the technical evidence digest. Students complete the same numbered tasks and rubric; technical failure does not lower the academic standard.

Formal lesson plan · Overview and alignment

Overview

Healthy lunar tomato plants flower but do not set fruit. Students use case evidence to model a process, identify its first failed event, and defend a diagnosis.

Audience

Middle school through adult/general audiences in environmental science, life science, agriculture, or STEM.

Prerequisite knowledge

Flowers are reproductive structures; fruits develop after successful reproductive events.

Standards position

This lesson **supports components of NGSS MS-LS1-4**; it does not claim full fulfillment merely because pollination appears in the case.

- **SEP:** Argument from Evidence; Developing and Using Models.
- **DCI:** LS1.B, plant reproduction and specialized structures/actions.
- **CCC:** Cause and Effect; Structure and Function.
- **Product:** Process model, diagnosis, CER, constrained design.

Measurable objectives

1. Sequence six pollination-to-fruit-set events.
2. Identify the first failed event with two observations.
3. Reject one alternative using evidence.
4. Write a compact CER.
5. Propose a support with criterion, constraint, and verification.

Success criteria

Distinguishes pollination, fertilization, and fruit set.	Locates failure at pollen release/agitation.
Uses undisturbed pollen plus an independent source.	Explains why vegetative health does not prove reproductive success.
Design restores the missing event and can be verified.	

Formal lesson plan · Procedure

0–5	**Setup** Open the game and packet. State that academic assessment uses the model and reasoning, not the game score.
5–10	**Launch** Present: “Plants are green and flowering, but no tomatoes form.” Students complete **2 · Initial thinking**. Do not mention wind, bees, or vibration.
10–28	**Investigation** Students obtain one clue from Crew, Sensors, Plants, and Logs. Prompt them to record evidence that locates a process step rather than transcribing dialogue.
28–38	**Evidence processing** Students complete **3 · Model the pollination sequence** and **4 · Identify the failed step**. Ask: “What is the last event that is definitely working?”
38–48	**Diagnosis and explanation** Students submit the game diagnosis, then complete **5 · Test the competing explanations**, **6 · Diagnose and reject an alternative**, and **7 · Claim-Evidence-Reasoning**.
48–55	**Application** Students complete **8 · Design a reliable pollination support**. Require criterion, constraint, and a biological success check.
55–60	**Exit** Students complete **9 · Exit ticket** independently. Collect or save the local fillable packet.

> **Optional mechanism test:** The stem-shaking interaction and hand/mechanical-pollination records can confirm the model, but exhaustive node visitation is never required.

Assessment, accessibility, and misconceptions

Checks for understanding

- What is the last confirmed working event?
- Is pollination the same event as fertilization?
- Which evidence rules out a general growth failure?
- What observation would prove a design restored the missing event?

Assessment

Use the quick or formal rubric. Evidence, mechanism, reasoning, alternative evaluation, vocabulary, and completion are academic targets.

Accessible participation

- Accessible role has larger type and linear modeling.
- Bullet-point evidence is acceptable before prose.
- All essential content is real text and keyboard reachable.
- Fallback digest preserves the same tasks.
- All work is individually completable.

Misconceptions to prevent

1. **Self-pollinating means no movement.** Pollen still must be released and reach the stigma.
2. **Pollination equals fertilization.** Fertilization occurs later.
3. **No bees means no reproduction.** Species and methods vary; diagnose the missing process, not an animal.
4. **Healthy leaves prove reproductive health.** Vegetative and reproductive success can separate.
5. **One numeric threshold is universal.** Game values are local case data.

Differentiation

Advanced students may analyze how unsuitable temperature or humidity could produce similar symptoms at a different step. Do not require extra work for full credit.

> SCIENCE BOUNDARY
>
> The established sequence supports the lesson. Shackleton personnel, readings, records, and exact engineering values are fictional. Do not present the game's fabricated technical brief as a real NASA source.

Teacher case analysis

Source	Clue ID	Evidence contribution
Botanist Chen	`FLOWERS_NO_FRUIT`	Healthy flowers drop without fruit.
Sensor Array	`LOW_AIRFLOW`	Air movement is nearly absent while major growth conditions are nominal.
Specimen Row	`POLLEN_UNDISTURBED`	Viable pollen remains on anthers; stigmas are clean.
Design Docs	`NO_POLLINATION_PLAN`	No animal, manual, or mechanical plan was implemented.

Correct mechanism

The last working event is viable pollen production. The first failed event is effective agitation/pollen release. Without stigma contact, pollen-tube growth, fertilization, and fruit set cannot proceed.

Essential evidence

Undisturbed pollen, still air/no insects, missing plan, and flower drop. Optional gated nodes strengthen but do not replace this chain.

Distractors

- Regolith toxicity
- Missing ultraviolet light
- Excess carbon dioxide

Each is weakened by healthy growth and the direct flower-level evidence.

Instructional opportunity

A direct intervention can test a causal model: agitating a flower releases pollen. This is stronger than memorizing a vocabulary answer.

Established
Flower structures and reproductive sequence.

Plausible extrapolation
Intentional support in an off-world habitat.

Fictional
Location, characters, readings, records.

Context-dependent
Best equipment and environmental limits.

Rubrics

Dimension	Secure	Developing	Beginning
Evidence	Specific evidence including undisturbed pollen and another source.	Some relevant evidence; bottleneck or diversity missing.	Vague, copied, or irrelevant evidence.
Mechanism	Correct sequence and first failure.	General pollination failure; chain incomplete.	Cause named without mechanism.
Reasoning	Evidence linked to claim; alternative rejected.	Connection partly explained.	Evidence listed without explanation.
Communication	Vocabulary accurate; required tasks complete.	Minor gaps.	Major gaps obscure reasoning.

Formal analytic dimensions

Process-model accuracy	Failed-step diagnosis
Evidence selection/diversity	Alternative evaluation
CER reasoning	Engineering response
Scientific qualification	Communication/completion

> **Completed exemplar:** Use the independent Answer Key role. Every keyable field in Tasks 3–9 is completed under Curriculum Bible v1.3.

Academic grading boundary

Game score, speed, rank, rapport, optional-node completion, and bonus insights are excluded. Teachers may apply a local numeric scale to the formal rubric, but no points appear on the Student page by default.

Technical fallback and references

FALLBACK PRINCIPLE

Technical failure must not become academic failure. Give the evidence digest and keep the same numbered tasks, expectations, and rubric.

Fallback evidence digest

A · Crew
Healthy flowers open, then drop without ovary swelling.

B · Sensors
Growth conditions are nominal; air movement is nearly absent and no fans are installed.

C · Plants
Viable pollen stays on anthers; stigmas are clean; gentle agitation releases pollen.

D · Records
Pollination remained “to be determined”; no system was installed or scheduled.

Authoritative references

1. University of Connecticut, “Pollination of Greenhouse Tomatoes,” revised 2024.
2. University of Alaska Fairbanks, “Pollination & Fruit Development in Tomatoes.”
3. UC Davis Tomato Genetics Resource Center, “Guidelines for Emasculating and Pollinating Tomato Flowers.”
4. University of Maryland Extension, “Tomato Pollination and Bumblebee Visits.”
5. Next Generation Science Standards, MS-LS1-4.

Figure-rights decision

Authoritative diagrams were researched but not copied or adapted because the reviewed pages did not provide an explicit open-content license. The process schematic is curriculum-original, editable, grayscale-first, and fully described in text.